KRISHNA ANVITH VATTIKUTI

AI / ML Engineer | LLM Systems, RAG, Agents, Evaluation

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SUMMARY

AI/ML Engineer (M.S. Data Science) who builds and deploys Large Language Model (LLM) applications: Retrieval-Augmented Generation (RAG), QLoRA/PEFT fine-tuning, tool-calling AI agents, guardrails, and LLM evaluation. Currently a Software Engineer at Enclavia AI, building the dashboard and metric engine for an on-premises AI agent platform, with a computer vision and time-series ML foundation in Python, PyTorch, and TensorFlow.

TECHNICAL EXPERIENCE

Enclavia AI - AI/ML Engineering Intern Sept 2026 – Present

- Built the dashboard specification and metric engine for an on-premises AI agent platform: a versioned widget contract, a DuckDB metric compiler with a SQL guard, and a semantic data model (column roles, status meanings, joins) that grounds agent-generated dashboards in customer data.
- Added LLM on-task relevance scoring to the guardrail pipeline, logged alongside guardrail decisions as a non-blocking signal for monitoring and evaluation.
- Prototyped wireframe-to-dashboard generation that renders uploaded HTML layouts, binds CSV data in place of sample figures, and runs section controls without executing untrusted scripts.
- Developed a spec-driven Vue dashboard designer with join validation and tests, YAML upload with automated data profiling.

UpGrad - Data Science Intern Jan 2024 – May 2024

- Developed a TensorFlow training pipeline for a Human Activity Recognition system across 6 activity classes, modularizing preprocessing and feature extraction into 4 stages to support faster model iteration.
- Applied ensemble classification and threshold tuning on accelerometer/gyroscope time-series data, reaching 97.5% accuracy and improving fall-detection reliability.
- Improved feature engineering and model iteration workflows, reducing preprocessing time by 30% and model-training time by 20%.

PROJECTS

AgentWatch - Observability & Evaluation Platform for AI Agents (Github-Repository) June 2026

- Built an observability and tracing platform for AI agents to monitor runs, inspect tool calls, debug failures, and evaluate output quality end-to-end.
- Developed a FastAPI backend, React + Vite dashboard, SQLite/SQLAlchemy storage layer, and a Python SDK to capture agent runs, spans, latency, inputs, outputs, statuses, and evaluation results.
- Implemented rule-based evaluations to automatically flag failed runs, slow responses, empty outputs, and tool-execution errors across simulated agent workflows — directly aligned with eval-driven LLM engineering practices.

RAFT - ArXiv Retrieval-Augmented Fine-Tuning (Github-Repository) March 2026

- Fine-tuned Phi-2 (2.7B) with QLoRA on a custom ArXiv QA dataset built from 200 papers to improve citation-grounded answer generation.
- Created a custom RAFT training dataset with an 80/20 oracle-to-distractor context ratio, teaching the model to cite relevant passages and refuse unsupported answers.
- Built a FastAPI side-by-side evaluation UI to benchmark RAFT vs. Llama-3 outputs on citation precision, answer faithfulness, and refusal accuracy.

MCP Agent Browser - LLM Browser Automation Agent (Github-Repository) Nov 2025

- Built an LLM-powered browser automation agent using LangChain ReAct planning and Playwright to convert natural-language instructions into executable actions (click, fill, navigate, extract).
- Validated the agent across 20+ real-world workflows and ran failure analysis on ambiguous instructions, prompt structure, tool execution, and multi-step planning.
- Improved agent reliability through iterative prompt refinement and documentation of workflow-level failure patterns, informing tool-calling and orchestration design.

TECHNICAL SKILLS

Generative AI & LLMs: LLMs, RAG, Fine-Tuning (QLoRA, LoRA, PEFT), Prompt Engineering, LangChain, Hugging Face Transformers, Vector Databases (ChromaDB), Ollama, Llama-3

AI Agents & Evaluation: Agentic AI, ReAct Agents, Model Context Protocol (MCP), Function Calling, Guardrails, LLM Evaluation, Observability, Playwright

Machine Learning: PyTorch, TensorFlow, TensorFlow Lite, Scikit-learn, Computer Vision, OpenCV, Time-Series, Feature Engineering, Model Quantization

Engineering & Data: Python, SQL, DuckDB, FastAPI, REST APIs, Docker, Git, CI/CD, Unit Testing, Vue, React, Pandas, NumPy, Tableau, Power BI

EDUCATION

New Jersey Institute of Technology - M.S. Data Science, Newark, NJ Aug 2024 – May 2026

Lovely Professional University - B.Tech Computer Science, Punjab, India Jul 2020 – Jul 2024